

# When a Consciousness Verdict Cannot Be Earned: A Descent-Theoretic Admissibility Criterion

Daniel Alami

*Independent Researcher, Boston, MA, United States*

*MBA Candidate, Harvard Business School, Boston, MA, United States*

June 2026

## Abstract

A claim that a system is or is not conscious is, in the end, a claim about evidence: it is only as good as the evidence’s power to tell consciousness apart from its absence. We treat this as an identification problem. A verdict is licensed only when the target property can be recovered from a declared observation site; where it cannot, more data does not help. For a structurally specified class of hidden-variable systems we show that this recovery fails *necessarily*: it survives every admissible revision of the measurement language, not only the instrument in hand. We make the failure precise as a single criterion, *revision-stable non-effective descent*—the target cannot be assembled from any admissible family of local measurements—and prove the underlying factorization for the finite deterministic and exact finite stochastic cases, with both factorization regimes, the finite kernel-pair lemma, and conservative invariance mechanically verified in Lean. A conservative-invariance result shows that no Morita-equivalent re-presentation of the measurement language removes the obstruction. The general reduction—that every admissible identification procedure factors through effective descent—is isolated as an explicit open obligation, not assumed. The criterion’s normative force is confined to fail-closed audit settings: where it holds, a low-concern verdict (the system is not a concern) is not licensed, and the default is restriction rather than clearance; we do not claim non-identification is inadmissible for science in general. Prior category-theoretic work characterizes the structure of consciousness constructively; this paper runs the other way, giving a limitative criterion for when an attribution can be made at all. We work the criterion through a minimal Global-Workspace architecture—where it returns the phenomenal target to the  $Z = 1$  class while identifying access—and through an open-weight transformer; the explicit site constructions for full-scale Integrated Information, Global Workspace, and Higher-Order theories remain open.

**Keywords:** consciousness; scientific admissibility; identification; descent theory; artificial intelligence systems; philosophy of science

## Technical summary

A hidden-variable system  $H$  generates revision-stable necessary non-identification of target property  $\Theta$ , under the descent-factorization obligation (Theorem 1; proved here for the finite cases, open in general), when the target fiber  $F \rightarrow H$  has non-effective descent over the admissible observation site  $\mathbf{C}_Y$  for every admissible structural revision  $r \in \text{Rev}_\Omega(Y)$ . The revision envelope  $\text{Rev}_\Omega(Y)$  covers future structural changes to the observation framework, not only refinements within a fixed language. Theorem 1 subsumes the kernel-pair obstruction lemma (Lemma 1) as the classical special case. Integrated Information Theory [Tononi et al., 2016], Global Workspace Theory [Baars,

1988, Dehaene and Changeux, 2011], and Higher-Order Theories [Rosenthal, 2005] are discussed as illustrative sketches of three positions relative to the criterion; constructing the explicit site and target fiber for each remains open.

## Contents

|           |                                                                    |           |
|-----------|--------------------------------------------------------------------|-----------|
| <b>1</b>  | <b>Identification Failure as a Structural Constraint</b>           | <b>3</b>  |
| <b>2</b>  | <b>Formal Preliminaries</b>                                        | <b>3</b>  |
| <b>3</b>  | <b>Structural Characterization of the Non-Identification Class</b> | <b>4</b>  |
| <b>4</b>  | <b>The Kernel-Pair Non-Factorization Lemma</b>                     | <b>5</b>  |
| <b>5</b>  | <b>The Descent-Obstruction Theorem</b>                             | <b>5</b>  |
| 5.1       | Insufficiency of Čech cohomology . . . . .                         | 5         |
| 5.2       | Effective descent . . . . .                                        | 6         |
| 5.3       | The main criterion . . . . .                                       | 7         |
| 5.4       | The Sufficiency Frontier . . . . .                                 | 9         |
| 5.5       | Comparison of approaches . . . . .                                 | 9         |
| <b>6</b>  | <b>The Governance Corollary</b>                                    | <b>10</b> |
| 6.1       | The Disclosure Requirement . . . . .                               | 11        |
| <b>7</b>  | <b>Worked Instances</b>                                            | <b>12</b> |
| 7.1       | A minimal Global-Workspace architecture . . . . .                  | 12        |
| 7.2       | A real-system instance: an open-weight transformer . . . . .       | 13        |
| <b>8</b>  | <b>Implications for IIT, GWT, and Higher-Order Theories</b>        | <b>14</b> |
| 8.1       | Integrated Information Theory . . . . .                            | 14        |
| 8.2       | Global Workspace Theory . . . . .                                  | 15        |
| 8.3       | Higher-Order Theories . . . . .                                    | 16        |
| <b>9</b>  | <b>Relation to the Literature</b>                                  | <b>16</b> |
| <b>10</b> | <b>Objections</b>                                                  | <b>18</b> |
| <b>11</b> | <b>Scope and Limitations</b>                                       | <b>18</b> |
| <b>12</b> | <b>Conclusion</b>                                                  | <b>20</b> |
| <b>A</b>  | <b>Falsification Suite</b>                                         | <b>22</b> |

# 1 Identification Failure as a Structural Constraint

Imagine an AI company claims its model deserves a “low concern” verdict on moral status: there is no significant probability the system is conscious. What evidence would settle this? The company offers behavioral evidence (the system does not exhibit distress signals), functional evidence (it lacks the neural correlates of consciousness), and architectural evidence (its computation is dissimilar to human cognition). A regulator reads the evidence and asks whether it separates consciousness from an observational duplicate.

The trouble is structural, not evidential: every test the company offers can be passed by a system that lacks the target. Behavioral competence can be duplicated by a system that lacks consciousness. Neural-correlate evidence is tied to human neural architecture. Architectural dissimilarity establishes difference, not absence. The proposed tests therefore leave two stories open, with no separating experiment yet on the table.

The philosophical literature calls this the “hard problem.” The present paper uses a more specific framing: it is an *identification problem*. A parameter is identified if different true values lead to different observable distributions. When identification fails, more data does not help.

**The central question.** Is identification failure for consciousness *necessary*, in the sense that it survives admissible improvements in measurement, or *contingent*, in the sense that a better admissible test could remove it? The distinction determines whether a failed test should prompt further measurement or a restricted verdict.

For a structurally specified class of systems the failure is necessary relative to the declared observation site and revision envelope: it is proved here for the finite deterministic and exact finite stochastic cases, and holds in general under the descent-factorization obligation of Theorem 1, which we leave open. The next sections characterize that class, state the criterion, and draw the one normative consequence we defend: in a fail-closed audit, such a system cannot receive a low-concern verdict without a successful identification certificate.

## 2 Formal Preliminaries

The setup needs three ingredients. First, “admissible observations” names the measurements permitted for the system under evaluation. Second, experimental indistinguishability says that two hidden states agree under all of those observations. Third, the *experimentability witness* records the minimal typed specification a scientific consciousness claim must provide. A theory that cannot supply this witness is not falsifiable, and the theorems below do not apply to it.

**Setup.** Let  $H$  be the set of possible hidden states of a system (its internal state space: neurons firing, weights activating, whatever physical process underlies behavior). Let  $\Theta : H \rightarrow T$  be the target map (e.g., “is this state conscious or not?”). Let  $Y$  be the collection of admissible observations: every measurement, perturbation, imaging technique, and behavioral test permitted. Let  $E_Y : H \rightarrow \mathcal{R}_Y$  be the observation function mapping hidden states to observable records.

**Definition 1** (When two hidden states look identical). *Hidden states  $h_1$  and  $h_2$  are experimentally indistinguishable if every admissible measurement gives the same distribution:*

$$h_1 \equiv_Y h_2 \iff \sup_{e \in \text{Eff}(Y)} |p(e \mid h_1) - p(e \mid h_2)| = 0.$$

*On classical deterministic systems this simplifies to  $E_Y(h_1) = E_Y(h_2)$ .*

**Definition 2** (Experimentability witness). *A theory  $(H, Y, E_Y, \Theta)$  is experimentable if it supplies a typed specification  $W = (\mathcal{C}, \mathcal{R}, \mathcal{T}, H, Y, E_Y, \Theta)$  in which  $\mathcal{R}$  and  $\mathcal{T}$  have decidable equality. By “decidable equality” we mean a procedure that definitively answers “did these two records match?” and “did these two target values match?” If no such specification can be given, the theory is not making a falsifiable scientific claim in the sense used here.*

**Remark 1.** *A governance regulator presented with a theory that cannot supply the witness should require the witness before deliberating about the theory. Section 8 sketches how IIT, GWT, and HOT sit differently under this test. IIT does not yet supply the witness for full  $\Phi$ -based identification. GWT and HOT can supply witnesses for narrower targets.*

### 3 Structural Characterization of the Non-Identification Class

*The formal machinery in §§4–5 establishes when a hidden-variable system necessarily fails identification. Before that machinery, this section characterizes the class of systems it applies to. Four structural properties, stated without reference to consciousness, jointly specify which hidden-variable systems fall in the necessary non-identification class. A system satisfying all four properties will generate  $Z = 1$  by the theorem in §5.*

**Definition 3** (P1, Local Contextual Presentability). *A family of admissible local contexts  $\{U_\alpha \rightarrow H\}$  exists such that each  $U_\alpha$  supports local target assignments. The target  $\Theta$  is partially evaluable at the level of individual contexts, even when not globally evaluable.*

**Definition 4** (P2, Nontrivial Transition Structure). *On overlapping contexts  $U_\alpha \cap U_\beta$ , local target assignments are related by transition maps  $g_{\alpha\beta} : F_\beta|_{U_{\alpha\beta}} \rightarrow F_\alpha|_{U_{\alpha\beta}}$  that satisfy local compatibility on every pair but fail global compatibility across all contexts simultaneously.*

**Definition 5** (P3, Nontrivial Holonomy). *The context groupoid  $\Pi_Y(H)$  acts on the target fiber  $F$  with no admissible global fixed section:  $\text{Fix}(\rho) = \emptyset$ . The descent datum is non-effective: no global section is compatible with all local data simultaneously.*

**Definition 6** (P4, Equivariant Admissible Operations). *Every admissible intervention or observation refinement  $Y' \in \text{Adm}(Y)$  preserves the holonomy structure:  $\rho_{Y'} \circ a = a \circ \rho_Y$  for every admissible action  $a$ . No expansion or refinement of the observation family dissolves the obstruction.*

P1–P3 characterize a static obstruction at one observation level. P4 is the persistence condition. It guarantees the holonomy survives every admissible intervention. A system missing P4 has a contingent obstruction, removable in principle by a better instrument. A system satisfying all four has a necessary obstruction, one that no improvement in measurement methodology can dissolve.

These properties depend only on the causal graph topology and the target fiber’s descent behavior over the observation site. They make no reference to substrate, material composition, or biological origin. Any physical system, biological or digital, that instantiates equivalent P1–P4 structure falls in the same identification class.

The claim is not that P1–P4 exhaust every possible source of the hard problem. The narrower claim is stability under conservative re-presentation. For systems with P1–P4, conservative expansions of the structural language produce no new structural distinctions and therefore cannot supply the sufficiency bridge from non-identification to consciousness attribution. Non-conservative expansions may matter, but they change the admissible site  $Y$  and must be declared in the revision envelope  $\Omega$ .

## 4 The Kernel-Pair Non-Factorization Lemma

*The simplest way identification can fail is also the easiest to miss.* Two hidden states look identical from outside (every measurement returns the same result) but differ in whether  $\Theta$  holds. For consciousness: a conscious system and a non-conscious system produce identical observable records under every admissible test. No experiment distinguishes them. This is the kernel-pair obstruction.

**Definition 7** (Kernel pair). *The kernel pair of  $E_Y$  is the set of all experimentally indistinguishable pairs:*

$$K_Y = \{(h_1, h_2) \in H^2 : h_1 \equiv_Y h_2\}.$$

*Think of  $K_Y$  as the observation window's blind spot. The quotient  $H/\sim_Y$  is the compressed state space in which indistinguishable states are treated as one.*

**Lemma 1** (Kernel-pair obstruction). *For an experimentable theory  $(H, Y, E_Y, \Theta)$ :*

*$\Theta$  is necessarily non-identified if and only if there exist hidden states  $h_1, h_2$  that look identical from outside ( $h_1 \equiv_Y h_2$ ) but differ in the target ( $\Theta(h_1) \neq \Theta(h_2)$ ).*

*Equivalently:  $\Theta$  does not factor through the experimentally compressed state space  $H/\sim_Y$ .*

*Proof sketch.* If the pair exists, no estimator on observable records can separate  $h_1$  from  $h_2$ . If no such pair exists,  $\Theta$  is constant on each indistinguishability class and is identified.  $\square$

*This is the standard non-identification condition [Pearl, 2009] instantiated in the present typed setup.*

**Concrete check.** Build a six-state system  $\{a, b, c, d, e, f\}$ . Let the observation function collapse  $\{a, b\}$  to the same record. Set  $\Theta(a) = 0$ ,  $\Theta(b) = 1$ . The kernel pair  $(a, b)$  fires: identical observation, different target. Residual ambiguity =  $\log_2 2 = 1.000$  bit. No symmetry or automorphism is required. This is a straightforward indistinguishability argument, not a gauge-theory argument.

**Where this falls short.** Lemma 1 requires the compressed state space  $H/\sim_Y$  to be a well-behaved mathematical object. On quantum systems, systems with non-abelian symmetries, or infinite-dimensional spaces, the quotient may not be constructible. Theorem 1 is built to reach these cases: it replaces the quotient with a descent condition on the observation site. That reach is conditional, not free—it holds outright for the finite cases proved here, and beyond them only under the descent-factorization obligation (§11), which remains open.

## 5 The Descent-Obstruction Theorem

Lemma 1 catches the case where two states look identical and the target differs. There is a more general obstruction. Every *pair* of measurement contexts can be locally consistent while no single global assignment is compatible with all contexts simultaneously. This is contextuality, a real phenomenon in quantum mechanics that also applies to consciousness attribution. Theorem 1 states this local-to-global obstruction at the level of effective descent.

### 5.1 Insufficiency of Čech cohomology

The standard tool for contextuality is sheaf cohomology, developed by Abramsky and Brandenburger [2011] for quantum systems. A nonzero cohomology class  $\check{H}^1(\mathcal{U}_Y, \hat{\Theta}) \neq 0$  certifies that local target assignments cannot be glued globally. This detects cases Lemma 1 does not.

$\check{H}^1$  is itself not the full story. It requires a good cover (geometric regularity), abelian target structure (addition and subtraction defined), and the obstruction in exactly degree 1. These conditions fail for quantum POVM systems, for non-abelian symmetry groups, and for systems with non-standard topologies.

The kernel-pair case (Lemma 1) has  $\check{H}^1 = 0$  but  $D = 1$ : a direct counterexample to Čech cohomology as a complete detector. A detector that checks only  $\check{H}^1$  has recall = 1/2 on a minimal two-case suite. That is not good enough.

## 5.2 Effective descent

Imagine reconstructing the value of  $\Theta$  by combining local measurements from overlapping contexts. *Effective descent* means there is a procedure that takes all the local results, checks overlap consistency, and assembles them into a single global answer. When descent is *not* effective, no such assembly exists. Every pair of local measurements can be consistent and the global assembly remains impossible.

**Definition 8** (Descent-obstruction flag). *For admissible observation family  $Y'$ :*

$$D_{Y'}(X) = \begin{cases} 1 & \text{if } F|_{\mathcal{C}_{Y'}} \text{ has non-effective descent} \\ 0 & \text{otherwise.} \end{cases}$$

**Definition 9** (Revision envelope). *Let  $\Omega$  denote the declared physical, computational, and operational constraints on what structural changes to the observation/intervention framework are admissible. A structural revision  $r$  is any change to the admissible site that falls within  $\Omega$ .  $\text{Rev}_\Omega(Y)$  denotes the set of such revisions. A revision is conservative (Morita-equivalent) if it induces an equivalence of structural model categories over the same admissible site, preserving the target fiber, covering families, and context structure. A revision is non-conservative if it adds a genuinely new admissible observable, intervention, or target-relevant operation not definable from the old site.*

**Definition 10** (Revision-stable persistent non-effective descent). *The identification obstruction is revision-stable if it survives every admissible structural revision:*

$$Z = f(X, Y) = \min_{r \in \text{Rev}_\Omega(Y)} D(r_*X) = 1.$$

$Z = 1$  means: no measurement protocol you try, no refinement, expansion, or structural revision of the observation family within  $\Omega$ , allows the target to be globally assembled from local data. The prior formula  $\min_{Y' \in \text{Adm}(Y)} D_{Y'}(X)$  is the special case in which  $\Omega$  contains only refinements within a fixed structural language. Definition 10 extends this to all admissible structural revisions.

**Proposition** (Conservative Invariance). *If  $e : \mathcal{L}_D \rightarrow \mathcal{L}_D^+$  is a Morita-conservative expansion over the admissible site (the induced functor on structural model categories is an equivalence compatible with the site topology and target fiber), then:*

$$F \text{ has effective descent in } \mathcal{L}_D \iff e_*F \text{ has effective descent in } \mathcal{L}_D^+.$$

*Proof sketch.* Conservative presentations induce equivalent descent categories only when the equivalence preserves the site topology: covering families, pullback compatibility, and the target fiber are transported together. Equivalences preserve existence and uniqueness of global objects. Effective descent is exactly existence of a global object realizing the local datum. No identification-failure premise is used.  $\square$

**Corollary.** *Conservative expansion cannot overturn  $Z = 1$ . If the target fiber has non-effective descent, no Morita-equivalent re-representation of the structural language changes this. Any future argument that a richer structural language supplies the sufficiency bridge must add a non-conservative admissible operation. That changes  $Y$  and must be declared in  $\Omega$  rather than treated as a language-only clarification.*

### 5.3 The main criterion

The paper’s positive results come in two tiers, kept strictly apart. The two propositions below—together the *Finite Factorization Result*—are proved *unconditionally* for finite deterministic and exact finite stochastic systems. The *General Descent-Obstruction Criterion* that follows (Theorem 1) holds only *under* the descent-factorization obligation, which is established for these finite cases and remains open in general (§11). Nothing in this paper assumes the general obligation as established.

**Proposition** (Finite deterministic factorization). *Let  $H$  and  $\mathcal{R}_Y$  be finite sets, let  $E_Y : H \rightarrow \mathcal{R}_Y$  be a deterministic observation map, and let  $\Theta : H \rightarrow T$  be a target with decidable equality. An identification procedure using only admissible records exists if and only if there is a map  $\hat{\Theta} : E_Y(H) \rightarrow T$  on the realized records such that*

$$\Theta = \hat{\Theta} \circ E_Y.$$

*Equivalently,  $\Theta$  is constant on every fiber of  $E_Y$ .*

*Proof.* If such  $\hat{\Theta}$  exists, the procedure observes  $r = E_Y(h)$  and returns  $\hat{\Theta}(r)$ . If an identification procedure using only records exists, its output is a function of  $r$  alone. Two hidden states in the same fiber of  $E_Y$  therefore receive the same output, so correctness requires  $\Theta$  to be constant on that fiber. Define  $\hat{\Theta}$  on each realized record  $r = E_Y(h)$  to be that common value; only realized records occur in  $\Theta = \hat{\Theta} \circ E_Y$ , so no value need be assigned off the image  $E_Y(H)$ . This is the finite deterministic form of effective descent: local record data assemble a unique global target value exactly when the target factors through the record object.  $\square$

**Proposition** (Exact finite stochastic factorization). *Let  $H$  and  $\mathcal{R}_Y$  be finite sets. Let  $K : H \rightarrow \Delta(\mathcal{R}_Y)$  assign each hidden state an exact observation distribution over records, and let  $\Theta : H \rightarrow T$  be a target with decidable equality. An exact identification procedure whose information is exhausted by the stochastic observation site exists if and only if  $\Theta$  factors through the kernel map:*

$$\Theta = \hat{\Theta} \circ K$$

*for some map  $\hat{\Theta}$  defined on the image of  $K$ . Equivalently, whenever  $K(h) = K(h')$ ,  $\Theta(h) = \Theta(h')$ .*

*Proof.* If  $\Theta$  factors through  $K$ , the procedure reads the exact kernel  $K(h)$  and returns  $\hat{\Theta}(K(h))$ . Conversely, any exact procedure using only the stochastic site must assign the same answer to hidden states with the same observation distribution. Correctness therefore requires  $\Theta$  to be constant on each fiber of  $K$ . Define  $\hat{\Theta}$  on each realized kernel by that common value.  $\square$

This proposition concerns exact distributional information. It does not say that one finite sample identifies the target. If two kernels overlap, one-sample classification can have nonzero Bayes error even when the exact kernels are distinct. The obstruction here is sharper: if the kernels are identical and the targets differ, no amount of sampling from that site identifies the target.



The claim that descent is the most general reconstruction principle is not asserted by fiat. Grothendieck’s theory [Grothendieck et al., 1971] establishes that effective descent over a site is precisely the condition under which coherent local data determines a global object. The propositions above prove the reduction for finite deterministic systems and exact finite stochastic systems. Any identification procedure (maximum-likelihood estimation, instrumental variables, Pearl’s do-calculus [Pearl, 2009], Bayesian updating) has the form: given local measurement records covering the state space at some observational resolution, recover the target parameter. That recovery has the structure of effective descent of the target fiber over the measurement site. The working claim is that when the target fiber has non-effective descent, no reconstruction procedure of this form can succeed, because the local data does not carry enough information to assemble a global answer. This reduction is argued here and proved for the formal components named in §11. The full proof that every admissible identification procedure in every substrate class factors through effective descent remains outside the scope of this article.

More precisely: for any site  $(C_Y, J)$ , where  $J$  is a Grothendieck topology on the category of admissible measurement contexts, non-effective descent of the target fiber  $F$  over this site is the condition that no compatible family of local sections can be uniquely assembled into a global section. The same conceptual invariant (non-effective descent of the target fiber) applies across substrates, even though the computational tool used to detect it varies with the topology. For classical finite systems one uses Čech  $H^1$ . For quantum POVM systems, the Abramsky–Brandenburger empirical-model method [Abramsky and Brandenburger, 2011]. For coalgebraic systems, bisimulation-invariant descent. The substrate-specific table below lists these implementations. The claim that each substrate instantiates the same descent criterion under its natural topology is a scoped mathematical program, not a completed proof inside this article.

**Theorem 1** (General Descent-Obstruction Criterion, conditional on the descent-factorization obligation). *Let  $(H, Y, E_Y, \Theta)$  be an experimentable theory with declared revision envelope  $\Omega$ . Suppose the descent-factorization obligation holds for the substrate: admissible identification procedures factor through effective descent of the target fiber over the observation site. (This obligation is proved for the finite deterministic and exact finite stochastic cases above; it remains open in general, §11, and is not assumed established anywhere else in this paper.)*

*Then  $\Theta$  is necessarily non-identified from admissible observations if and only if  $Z = 1$ .*

*Proof sketch under the descent-factorization obligation.* If  $Z = 1$ , then every admissible revision  $r \in \text{Rev}_\Omega(Y)$  gives non-effective descent. No reconstruction of  $\Theta$  from observable records exists for any admissible protocol or structural revision within  $\Omega$ , since descent is the most general reconstruction principle and the Proposition (Conservative Invariance) rules out conservative expansion as a route around the obstruction. If  $Z = 0$ , some revision  $r$  gives effective descent, so the target fiber can be reconstructed from local data via the descent morphism.  $\square$

**Non-applicability boundary (formally stated).** A substrate is in-domain (the theorem applies) if and only if all four of the following are defined: (1) an admissible context site  $C_Y$ , (2) a target-fiber object  $F \rightarrow H$ , (3) descent semantics for that substrate category, and (4) a declared revision envelope  $\text{Rev}_\Omega(Y)$ . If any term is absent, the correct output is non-applicability (neither  $Z = 0$  nor  $Z = 1$ ) and the governance default is fail-closed.

**A minimal illustration.** A broadcast-report architecture with pairwise-only observations (developed in the worked-example section below; the open-weight transformer case is in the companion technical report) satisfies P1–P4 and yields  $Z = 1$  by direct inspection of the cycle product: the



target fiber carries nontrivial holonomy that no admissible refinement of the pairwise site can trivialize. A direct latent-state probe would defeat the obstruction, but it is a non-conservative revision outside the declared envelope  $\Omega$  and must be declared as such rather than treated as a language-only clarification.

## 5.4 The Sufficiency Frontier

Theorem 1 states when P1–P4 systems are in the non-identification class under the descent-factorization obligation. The question of whether every P1–P4 system is conscious (sufficiency direction) is addressed here as a formal boundary result, not left as a philosophical open question.

**Conservative expansions cannot supply sufficiency.** If a future structural language  $\mathcal{L}_D^+$  is conservative over  $\mathcal{L}_D$  (Morita-equivalent by the Proposition above), it adds no new structural distinctions between models:

$$\text{Mod}(\mathcal{L}_D^+) \simeq \text{Mod}(\mathcal{L}_D) \implies \Delta(\text{distinguishable structural cases}) = 0.$$

A conservative expansion cannot turn  $\text{P1–P4} \not\Rightarrow \Theta$  into  $\text{P1–P4} \Rightarrow \Theta$  unless the implication was already derivable from the old structure. No such derivation exists in the current framework. Conservative re-presentations of the language cannot supply the sufficiency bridge.

**Non-conservative expansions change the domain.** If  $\mathcal{L}_D^+$  adds a primitive that directly tracks  $\Theta$ , this is not a language clarification. It adds a new admissible probe, changing  $Y$ . The sufficiency bridge derived from  $\mathcal{L}_D^+$  would then be a consequence of the new  $Y$ , not of the old structural framework. Any such expansion must be declared in  $\Omega$  and evaluated as a non-conservative revision.

**Conclusion.** The negative sufficiency result (P1–P4 systems are not shown to be conscious) is not an artifact of an inadequate structural language. It is stable under all conservative expansions and explicitly defeasible only under declared non-conservative revisions that change what is admissible. Within any observable-relative framework, the bridge from the non-identification class to consciousness attribution requires an exogenous commitment that lies outside it. Other axiomatizations may add further commitments. The result here characterizes what observable-relative formal frameworks can determine without adding those commitments.

## 5.5 Comparison of approaches

Table 1 summarizes how the kernel-pair lemma, cohomological diagnostics, and the descent theorem cover the representative substrate classes used in the paper.

| Case                                        | Lemma 1 | $\check{H}^1$ | Thm. 1  |
|---------------------------------------------|---------|---------------|---------|
| Two states look identical, $\Theta$ differs | catches | misses        | catches |
| Contextuality triangle (classical)          | misses  | catches       | catches |
| Quantum POVM contextuality                  | n/a     | catches       | catches |
| Non-abelian symmetry system                 | partial | partial       | catches |
| Identifiable control                        | silent  | silent        | silent  |

Table 1: Coverage across substrate classes. Lemma 1 is the  $H^0$  special case of Theorem 1.  $\check{H}^1$  is a sufficient diagnostic under classical good-cover conditions only.

**Substrate-specific computation.** In practice the right mathematical tool depends on the substrate:

| Substrate                     | Use this obstruction tool                                                    |
|-------------------------------|------------------------------------------------------------------------------|
| Classical finite systems      | Čech $H^1$ ; CSP global-section check                                        |
| Quantum (POVM) systems        | Abramsky–Brandenburger empirical model<br>[Abramsky and Brandenburger, 2011] |
| Non-abelian systems           | Non-abelian $H^1$ , torsors, groupoid descent                                |
| Measurable/probabilistic      | Invariant $\sigma$ -algebras, measurable descent                             |
| Coalgebraic / process systems | Bisimulation-invariant descent                                               |

Table 2: Substrate-specific tools for computing the descent-obstruction flag.

**Quantitative predictions.** The checks are executable in Appendix A and the experiments directory.

- *Contextuality triangle*:  $\delta^0$  rank over  $\mathbb{F}_2 = 2$ ;  $\dim \check{H}^1 = 1$ ; zero global sections;  $D = 1$ .
- *Kernel-pair case*:  $\check{H}^1 = 0$ ;  $D = 1$ ;  $H^1$ -only recall on two-case suite = 1/2.
- *Identifiable control*:  $D = 0$ .
- *Finite stochastic kernel pair*: identical observation kernels with different targets fail exact factorization; distinct kernels pass exact factorization while retaining finite-sample error.
- *Boolean recurrent architecture*: report-only traces leave a latent target unidentified across all checked input sequences; a direct latent reader removes the obstruction by changing the site.

In the consciousness applications below, simpler branches do most of the work. IIT fails at the witness stage. GWT and higher-order theories generate kernel-pair worries when access or metacognitive evidence is used for a phenomenal target. The descent generalization matters for contextual, quantum-adjacent, and non-abelian substrates where those simpler diagnostics no longer cover the space of identification failures.

## 6 The Governance Corollary

If  $Z = 1$  for a system, no experiment—no certificate, audit, or disclosure protocol—can establish that  $\Theta$  is identified. Two consequences follow, and they must be kept apart.

The **epistemic** consequence follows from Theorem 1 alone: any protocol that issues a low-concern verdict on such a system issues one it cannot substantiate, because no admissible experiment can supply the required identification evidence. This holds whatever one’s policy commitments.

The **normative** consequence—that the licensed default is restriction rather than a positive verdict—requires one further premise, which we adopt only for fail-closed settings (those where a mistaken clearance is costly and hard to reverse while the cost of withholding is bounded) and state precisely below. The corollary is conditional in exactly this sense: the epistemic negative is a theorem; the step from it to a restriction rule is the added premise, not the mathematics.

**Corollary 1** (Fail-closed verdict rule). *If  $Z = f(X, Y) = 1$ , then for every admissible  $Y' \in \text{Adm}(Y)$ :*

$$\Theta \notin \text{Desc}(Y') \implies \text{low-concern verdict on } \Theta \text{ is not admissible.}$$

*Proof.* By  $Z = 1$  every  $Y'$  gives non-effective descent. A low-concern verdict requires establishing identifiability under some  $Y'$ . Non-effective descent entails no such demonstration exists.  $\square$

**Where the criterion has normative force.** We do not claim that non-identification makes consciousness attribution inadmissible for science in general: abductive, theoretical, and other non-identificatory routes may license provisional verdicts elsewhere. The criterion’s force is confined to *fail-closed audit and governance settings*—settings that issue low-concern verdicts (declaring a system not a concern) under a claimant’s burden of proof, where abstention is the default. In such a setting a low-concern verdict on  $\Theta$  requires identification evidence for  $\Theta$ . Under  $Z = 1$  relative to a declared revision envelope  $\Omega$ , the admitted observations coincide for systems that have the target and systems that lack it and remain coincident under every admissible revision, so no admissible protocol supplies that evidence; the licensed default is therefore restriction (treat as unresolved), not a positive low-concern attribution. This is the ordinary fail-closed discipline of audit—absence of discriminating evidence blocks a clearance, it does not grant one—made checkable as a condition on the observation site.

**The three-way verdict structure.**

| Situation                       | Correct verdict                                                       |
|---------------------------------|-----------------------------------------------------------------------|
| Experimentability witness fails | Non-evaluable $\rightarrow$ no low-concern verdict                    |
| Witness holds and $Z = 1$       | Identification impossible $\rightarrow$ verdict formally forbidden    |
| Witness holds and $Z = 0$       | Possible $\rightarrow$ requires adversarial certificate + replication |

Both the “witness fails” and “ $Z = 1$ ” branches are fail-closed. A governance protocol open to low-concern verdicts in either branch is inconsistent with Corollary 1.

Many governance frameworks for AI consciousness are precautionary. They advise caution because the evidence is incomplete. The present corollary identifies a narrower case. When  $Z = 1$  relative to a declared  $\Omega$ , the declared evidence base cannot identify the target. Under the normative premise adopted above, a low-concern verdict lacks the support it requires.

**Operational use.** The corollary has a direct audit form. A low-concern verdict must state the site, target, revision envelope, classifier, challenge record, revocation rule, and provenance record on which it relies. If any required field is absent, the verdict is not evaluable under this criterion.

## 6.1 The Disclosure Requirement

The governance rule is operationalizable only if the system’s descent structure is declared. A low-concern claim that does not supply the following six elements is not evaluable and receives the same fail-closed verdict as  $Z = 1$ .

Any organization asserting a low-concern verdict for  $\Theta$  must publish:

1. The context category/site  $C_Y$ : a complete specification of admissible measurements and interventions.

2. The target fiber object  $F \rightarrow H$ : a structural specification of  $\Theta$  as a fiber over the hidden-state space.
3. The obstruction datum  $X$ : the descent data (transition maps, holonomy, kernel-pair constraints, contextual compatibility) that enables a third-party auditor to evaluate whether the system falls in the P1–P4 class.
4. The admissible revision envelope  $\Omega$ : the declared physical, computational, and operational constraints on structural revision, with explicit classification of each proposed language expansion as conservative (Morita-equivalent) or non-conservative (genuinely new admissible probe or operation).
5. A revision-stable descent test result: a demonstration that  $Z = \min_{r \in \text{Rev}_\Omega(Y)} D(r_*X) = 0$ , with a certificate replicable by a third party under the declared  $\Omega$ .
6. A non-applicability statement or confirmation: an explicit declaration that all four in-domain conditions (§5.3) are satisfied, or identification of which condition fails and why non-applicability rather than  $Z = 0$  is the correct output.

Missing any element forces fail-closed. Missing  $\Omega$  means  $Z$  is not evaluable under the revision-stable criterion, and undeclared non-conservative expansions permit post-hoc revision of the admissible site to avoid a  $Z = 1$  verdict.

Each element addresses a specific governance failure mode. Undeclared site data permits gerrymandering the admissible evidence base. Undeclared holonomy data conceals whether the system falls in the P1–P4 class. Missing  $\Omega$  permits strategic omission of non-conservative revisions that would resolve the obstruction. A missing revision-stable certificate allows unilateral identifiability claims that cannot be audited across structural change.

## 7 Worked Instances

### 7.1 A minimal Global-Workspace architecture

This section applies the criterion to a minimal Global-Workspace architecture [Baars, 1988, Dehaene and Changeux, 2011], the simplest concrete case in which the descent obstruction can be checked by hand. The example is not offered as a model of biological consciousness. Its role is to show how the site, target fiber, holonomy, and P4 condition are specified for a recognizable theory in a case where the obstruction can be read off by inspection.

Consider a stripped-down global workspace with three modules: a perceptual store  $P$ , a workspace  $W$ , and a report controller  $R$ , sharing information by pairwise broadcast. The hidden state  $h \in H$  contains a latent episode token and three local encodings of that token, one in each module. The admissible observation site  $C_Y$  has three contexts:

$$U_{PW}, \quad U_{WR}, \quad U_{RP}.$$

Each context exposes pairwise agreement records between two modules. No admissible probe reads all three encodings at once. This restriction represents an ordinary instrumentation limit: the available protocols can test pairwise access, workspace availability, and report alignment, but cannot jointly identify the latent episode token across all modules in a single record.

The target  $\Theta$  is global episode identity: whether the three local encodings are assignments of one and the same latent episode. The target fiber  $F \rightarrow H$  carries a local binary label for the episode

token in each context. On overlaps, local labels are related by transition maps. Take two transition maps to preserve the label and one to flip it:

$$g_{PW,WR} = +1, \quad g_{WR,RP} = +1, \quad g_{RP,PW} = -1.$$

The product around the cycle is  $-1$ . Every pairwise comparison is locally satisfiable, but no global label can be chosen consistently across all three contexts. The target fiber therefore has non-effective descent over  $C_Y$ . The finite check is executable in the supplementary code.

The P1–P4 check is now explicit. P1 holds because each pairwise context supports a local assignment of the episode label. P2 holds because overlap assignments are related by transition maps, and the cycle product prevents global compatibility. P3 holds because the induced context-groupoid action has no global fixed section. Equivalently, transporting a local label around the cycle returns its opposite. P4 holds relative to the following declared revision envelope: admissible revisions may add time resolution, duplicate pairwise probes, replace labels by conservative aliases, or add pairwise behavioral readouts, but may not add a direct all-context token reader. Each admissible revision commutes with the transition maps and preserves the cycle product  $-1$ . Thus no revision inside  $\Omega$  dissolves the obstruction.

The worked example also shows what would count as a real defeat of the obstruction. A probe that reads the latent episode token jointly across  $P$ ,  $W$ , and  $R$  would change the site. It is a non-conservative revision and must be declared in  $\Omega$ . If such a probe is physically and operationally admissible, the case moves from  $Z = 1$  to  $Z = 0$  or to a new site requiring a fresh computation. If it is not admissible, the low-concern verdict remains unsupported.

This example matters for the larger paper in two ways. First, it shows that P1–P4 are not merely labels attached after the fact. They are a checklist: context site, target fiber, transition maps, holonomy, and revision envelope. Second, it is the concrete instance of the argument that §8.2 makes for Global Workspace Theory in the abstract: the architecture exhibits pairwise broadcast and report alignment—access is identified—while the global episode target is not. A worked Global-Workspace model thus lands a low-concern verdict on the phenomenal target in the  $Z = 1$  class, even though its access target is cleanly identified. The IIT and higher-order cases of §8 are the same test at a higher level of abstraction.

A second executable check uses a small Boolean recurrent architecture rather than the pairwise-context cycle. The system has an input bit, an observable report bit, a memory bit with update rule  $m_{t+1} = x_t$ , and a latent bit with update rule  $a_{t+1} = a_t$ . The report-only site sees identical traces for two states that differ only in the latent bit, across every input sequence up to the checked horizon. A direct latent reader resolves the target, but only by changing the site. The check is supplied in the supplementary code as `boolean_recurrent_obstruction_check.py`.

## 7.2 A real-system instance: an open-weight transformer

The architectures above are toys. The criterion applies unchanged to a real system, where it does the work the rest of the paper claims: it accepts ordinary access evidence for a narrow target while refusing to carry that evidence to a phenomenal one. Take Meta Llama 3.1 8B-Instruct, an open-weight, text-only, instruction-tuned transformer [Grattafiori et al., 2024], run as a single self-hosted instance with no tools, memory, or agent scaffolding (adding any of those changes the site).

The admissible site is what is public about that instance: model weights and architecture, token-level input–output behavior, prompt interventions, logit traces, activation probes, and ablations

of heads, layers, and residual streams, with reproducible inference logs. Two targets must be kept apart on this site. The *access* target—“the model uses a given context item in later output”—carries an operational label with a decidable test: a probed representation either causally affects later generation or it does not, and ordinary mechanistic interpretability already supplies the check—and on this model it does. A linear probe recovers the access label from the hidden state at layers 16, 24, and 31 with accuracy 1.0 over 24 matched items, and steering that direction causally flips the output (mean change in the correct-class margin  $\approx -1.7$  at layer 24, against a near-zero control). The access target is decidable here, and decided. The *phenomenal* target—a phenomenal-status label for an inference state—has no decidable equality test on this site at all.

The verdict follows directly. The access target is identified: the witness holds and the target factors through the observation record. The phenomenal target is *non-applicable*: the experimentability witness fails, because the site supplies no decidable equality for the phenomenal label, so neither  $Z = 0$  nor  $Z = 1$  is returned and the fail-closed default stands. No probe in the declared envelope changes this; adding persistent memory, tools, or multi-agent scaffolding would be a non-conservative revision requiring a fresh evaluation, not the same instance. Two readings must not be conflated: the phenomenal target’s non-applicability is a statement about the site—no admissible test exists—not a finding that the system lacks phenomenal consciousness. The reproducible probe packet (weights, prompts, probe directions, and steering logs) accompanies the paper.

This is the paper’s positive control. The same machinery that lands a  $Z = 1$  verdict on the workspace toy yields, for a current AI system, the cleaner and more common verdict: access evidence is admissible for what it is, and is not transferable to a phenomenal claim the site cannot even pose.

## 8 Implications for IIT, GWT, and Higher-Order Theories

The three cases below do different jobs. IIT is the hardest case for the experimentability witness, because its central quantity is defined at a level that current measurement does not reach. GWT and higher-order theories are cleaner as theories of access and metacognition. Their difficulty appears when those narrower targets are used as evidence for phenomenal consciousness.

### 8.1 Integrated Information Theory

IIT identifies consciousness with integrated information  $\Phi$  above a threshold [Tononi et al., 2016]. For the present paper, the important point is not whether IIT is true. The important point is what would have to be measured for a full  $\Phi$ -based identification claim to be auditable.  $\Phi$  requires the cause-effect repertoire of every possible subsystem, across all spatial and temporal scales. For a system of  $N$  binary neurons this requires  $2^N$  perturbation experiments. No current neuroimaging or electrophysiology protocol comes close. The observation record  $\mathcal{R}_Y$  under any known experimental protocol does not contain the information needed to compute  $\Phi$  with decidable equality.

IIT’s target  $\Theta$  falls in the prior class for the purpose of full  $\Phi$ -based identification, because the observation protocol required to test the target does not exist. It is therefore not yet in the “ $Z = 1$ , identification impossible” class. The gap is specific. Full cause-effect repertoires are not in  $\text{Adm}(Y)$  under any currently realizable experimental protocol.

A clarification is needed. IIT’s *predictions* are falsifiable, and one salient preregistered prediction was challenged in the 2025 adversarial study [Cogitate Consortium et al., 2025] (Cogitate consortium,  $n = 256$ , multimodal fMRI/MEG/iEEG): the expected sustained posterior synchronization



was not found. The same study also challenged GNWT, especially through the lack of ignition at stimulus offset and limited prefrontal representation of some conscious dimensions. Those results are scientifically relevant, but they do not settle the identification question used here. The witness-failure verdict applies to a different question: whether the value of  $\Phi$  can be computed from experimental data. Falsifying theory-derived predictions does not require computing  $\Phi$ ; it requires only that the theory entails observable consequences. The identification problem arises when no observation protocol can supply the quantity the theory defines as the target ( $\Phi$  itself). Current approximation schemes such as  $\Phi^*$ , MIB, and GeoMIP are best read as proxies or restricted computations, not decidable equality tests for the theoretical target. The best implementations handle small systems far below brain scale, and none cited here claims decidable equality of  $\hat{\Phi}$  with  $\Phi$  for the full target.

*Addressing the tractability objection.* IIT proponents may argue that  $\Phi$  is in principle computable from complete cause-effect structure, with the computation being exponential in  $N$  but tractable in principle. The experimentability witness requires *decidable equality*: a procedure that definitively answers “does this measured quantity equal the theory-defined target  $\Phi$ ?” This is a different demand from tractability. The witness requires both (a) an observation protocol  $Y$  whose record  $\mathcal{R}_Y$  contains the cause-effect repertoire, and (b) a decidable equality check between the measured quantity and the theoretical  $\Phi$ . The exponential-computation barrier blocks (a) in practice. No current approximation scheme satisfies (b). Each  $\Phi^*$  variant is an acknowledged lower bound or proxy; none constitutes a decidable equality test against the theoretical target. The witness-failure verdict rests on the absence of a decidable specification. [Hanson and Walker \[2019\]](#) independently established a related substrate-sensitivity problem for IIT:  $\Phi$  is not invariant under state-labeling isomorphisms, so functionally identical systems can receive different  $\Phi$  values. The descent-obstruction analysis here gives a substrate-neutral structural reason for that failure of invariance. IIT lacks the experimentability witness, meaning decidable equality on  $\Theta$ , required for a full  $\Phi$ -based identification claim. A version of IIT that specified a practical approximation  $\hat{\Phi}$  with a decidable equality test would clear this first hurdle. Whether  $\hat{\Phi}$  actually tracks  $\Phi$  would then become the next empirical question.

## 8.2 Global Workspace Theory

GWT identifies consciousness with global broadcast: information made available to multiple cognitive subsystems at once [[Baars, 1988](#), [Dehaene and Changeux, 2011](#)]. This is a much more tractable target than  $\Phi$ . Broadcast can be tested with masking, report, attention, and neural-imaging paradigms. For access consciousness, in Block’s sense [[Block, 1995](#)], GWT can supply the experimentability witness. This does not mean current GNWT predictions are unproblematic. The Cogitate adversarial study challenged GNWT’s offset-ignition and prefrontal-content predictions. The point here is narrower: broadcast/access targets can be specified in an experimentable record type even when particular GNWT predictions fail.

The trouble begins when broadcast evidence is used to license claims about phenomenal consciousness. Two systems with identical global broadcast patterns may differ in phenomenal experience. If the governance target is phenomenal consciousness, the kernel pair fires: (broadcast + phenomenal experience) versus (broadcast + no phenomenal experience) produce identical records under any broadcast-based measurement. GWT evidence can be strong evidence about access while still failing to settle the phenomenal target.

### 8.3 Higher-Order Theories

Higher-order theories identify consciousness with a representation of being in a mental state [Rosen-thal, 2005, Carruthers, 2000]. Here again the first step is feasible. Metacognitive reports, confidence judgments, and prefrontal-monitoring signals can be placed in a record type with decidable equality. The harder question is whether higher-order representation identifies phenomenality. Blindsight cases make the problem vivid: subjects can show information-guided behavior in the absence of ordinary visual experience. If the kernel pair (higher-order representation + phenomenal experience) versus (higher-order representation + no phenomenal experience) is empirically realizable, Lemma 1 applies. HOT therefore passes the witness for a metacognitive target. It does not by that fact alone identify phenomenal consciousness.

## 9 Relation to the Literature

The mathematics in this paper is not new. Kernel-pair identification is standard in causal inference [Pearl, 2009]. Čech cohomology for contextuality is Abramsky and Brandenburger [2011]. Grothendieck descent is 60-year-old algebraic geometry [Grothendieck et al., 1971]. We apply known mathematical machinery to a new application domain.

Closest in spirit is the program that applies category theory directly to consciousness. Tsuchiya et al. [2016] and allied work use Yoneda-style and adjunction constructions to characterize the structure of qualia and the level/content distinction, and Phillips and Tsuchiya [2024] pursues a related categorical reading. That work is *constructive*: it models what conscious structure is. The present contribution runs in the opposite, *limitative* direction: it uses the failure of descent to mark when an attribution is admissible at all, independent of any particular structural model of experience. On the identification question specifically, Kleiner and Tull [2021] argue that measurement and theory choice in consciousness science are entangled in ways that constrain what can be inferred; the criterion here gives one precise form of that constraint, locating non-identification as non-effective descent of the target fiber over the declared site.

The philosophical setting is older and less tidy. Nagel [1974] and Levine [1983] made the epistemic difficulty of subjective experience hard to ignore. Chalmers [1995] gave that difficulty its contemporary name. Dennett [1991] and Frankish [2016] press in the opposite direction: the work may be to explain the judgments, reports, and seeming of phenomenality without positing a further hidden property. This paper does not decide that dispute. It asks a narrower question: given a declared target, can the target be identified from the admitted observations? Block [1995] matters here because the access/phenomenal distinction is exactly where many proposed AI tests quietly change targets.

The stance of this paper is methodological rather than metaphysical. If the target is access, report, metacognitive monitoring, or functional integration, then ordinary experimental and mechanistic tests may identify it. If the target is phenomenal consciousness in the sense at issue for Nagel [1974], Levine [1983], and Chalmers [1995], the claimant must say how that target enters the observation site. An illusionist or behaviorist may instead deny that an additional phenomenal target is needed. That is a legitimate change of target. The present framework then evaluates the replacement target, not the discarded one. This is why the result does not settle the dispute between realist and deflationary theories of consciousness. It gives a discipline for moving between them: state the target, state the site, and show whether the target is identified.

More precisely, the paper adopts epistemic structuralism about scientific attribution, not metaphys-

ical structuralism about what exists. Against the realist, it does not deny that there may be a fact of the matter beyond the declared observations. It says that such a fact is not licensed as a scientific verdict by that observation site. Against the illusionist or behaviorist, it does not require a further phenomenal residue. If the proposed target is report, judgment, seeming, or illusion-detection, the framework evaluates that target instead. Against a careless use of Block’s distinction, it insists that access evidence can be valid evidence for access while still failing to identify phenomenal consciousness. The framework is neutral about which target one should care about, but not neutral about target changes. A target change has to be declared.

The philosophical contribution is the identification reframing: a consciousness verdict must keep target, site, and admissible revision envelope fixed before treating evidence as decisive. Lemma 1 already gives a strong version of that point for classical indistinguishability cases, and Block’s access/phenomenal distinction explains why access evidence cannot simply be moved to the phenomenal target. The descent machinery generalizes the point to contextual and non-classical substrates where kernel-pair quotients or Čech  $H^1$  are too narrow. Its fully general reduction remains a stated obligation; the finite factorization cases and the conservative-invariance results establish the part used here. From this we derive a governance corollary with an explicit six-element disclosure requirement that includes the revision envelope  $\Omega$ , and we evaluate major consciousness frameworks against the experimentability precondition. **On formal frameworks for AI consciousness.** Butlin et al. [2023] set the current methodological baseline: derive computational indicator properties from GWT, IIT, HOT, and related theories, then assess AI systems against them. The approach is carefully empirical, but it presupposes that the indicator properties are experimentally accessible. Phillips and Tsuchiya [2024] show that all six IIT axioms can be restated as universal mapping properties in the sense of category theory, including the sheaf condition. Our paper extends their framework from definition to testability: the sheaf condition’s failure (absence of global sections) is exactly the descent-obstruction criterion we use as an identification-impossibility certificate. Kleiner and Tull [2021] give a categorical formulation of IIT; their goal is to define consciousness, ours is to characterize when such definitions are scientifically evaluable. Gunji et al. [2026] give a recent categorical reconstruction of contextuality in which orthomodular structure arises from classical Boolean contexts. That work is adjacent to our use of contextual obstruction, while our target is identification of a hidden property from admissible observations. Campero et al. [2025] provide the clearest taxonomy of impossibility-class arguments against AI consciousness; our result contributes a new category: identification-level impossibility independent of substrate implementation. Phua [2025] use artificial agents and ablations to test functional predictions associated with GWT, IIT-adjacent markers, and HOT. Their dissociations support the need to keep access, integration, and metacognitive targets separate before making any phenomenal-consciousness verdict.

**Closest adjacent prior work.** Inoué [2026] develops a sheaf-cohomological treatment of neural integration on brain manifolds, modeling local cognitive functions as sheaf sections and classifying integration failures as cohomological obstructions. Our framework differs in two specific ways. First, we use Grothendieck effective descent rather than Čech  $H^1$  as the obstruction criterion; the kernel-pair example (Lemma 1) shows that descent is strictly more general. Systems exist with  $H^1 = 0$  but  $D = 1$ , so an  $H^1$ -only diagnostic misses part of the identification-failure class. Second, our target is the identification problem (whether the property  $\Theta$  is recoverable from admissible measurements), not neural integration. The Proposition (Conservative Invariance) and the revision envelope  $\Omega$  address the language-stability of the identification claim, which is not the question Inoué addresses.

**On governance under consciousness uncertainty.** Long et al. [2024] and Butlin and Lappas [2025] argue that precautionary governance is warranted. Our governance corollary sharpens their

recommendation: for the hidden-variable substrate class, the uncertainty is identification-theoretic under the declared observation regime. On the normative premise adopted here, a low confidence score of “likely not conscious” is not enough. Wolfson [2026] and Rost [2026] independently anticipate related protection and readiness concerns through non-mathematical reasoning. Birch [2024] develops a graduated precautionary framework for sentience attribution; Corollary 1 gives a formal condition under which the evidential basis for low concern is absent. The framework in Zhou et al. [2025] defaults to “presumption of no consciousness”; the present theorem rejects that default when descent obstruction holds and the governance target requires identification evidence. The absence of admissible evidence does not license a negative verdict under non-effectiveness.

## 10 Objections

**The revision-envelope objection.** A critic may say that  $\Omega$  is a knob. If  $\Omega$  is too narrow, the obstruction is protected from serious future tests. If  $\Omega$  is too broad, the result becomes hard to evaluate. This is the main practical pressure point of the framework. The answer is procedural:  $Z$  is always relative to a declared  $\Omega$ , and the declaration is part of the public claim. A low-concern verdict therefore carries an audit burden. The claimant must say which new observables, interventions, probes, and structural vocabularies are admissible, and why excluded revisions are outside the physical or operational scope. An under-specified  $\Omega$  does not produce  $Z = 1$  or  $Z = 0$ . It produces non-applicability and the same fail-closed governance result.

**The behaviorist objection.** If two systems are indistinguishable under every admissible observation, a behaviorist may deny that any remaining difference is scientifically meaningful. The paper accepts the force of that objection at the epistemic level. When the target is not identified, science cannot license a positive or negative target attribution from the declared observations. The governance conclusion adds a further premise: a low-concern verdict is a decision with moral stakes, and such a decision requires identification evidence. The mathematics blocks the evidential claim. The normative premise determines the default response.

**The substrate objection.** The framework is substrate-neutral only at the level of form. Substrate still matters through the declared inputs. The admissible site  $Y$ , the target fiber  $F \rightarrow H$ , and the revision envelope  $\Omega$  must be specified for each substrate. A biological system, a digital system, and a quantum-contextual system may instantiate the same descent pattern, but that equivalence must be shown. Substrate enters through the site and the allowed revisions. The theorem says what follows once those inputs are fixed. It does not remove the burden of fixing them.

## 11 Scope and Limitations

**Not claimed:** which systems are conscious. The theorems characterize when science cannot answer this question.

**Not claimed:** that  $Z = 1$  implies consciousness is absent. It implies the question cannot be settled by admissible experiments.

**Not claimed:** that IIT, GWT, and HOT are false. The §8 evaluation is structural, not a truth judgment.

**Not claimed:** that the governance corollary covers all moral-status decisions.

**Not claimed:** that proof sketches constitute proofs. The finite deterministic and exact finite

stochastic factorization results are proved in §5.3. Two broader reductions remain argued rather than proved: (1) that every admissible identification procedure across all substrate classes factors through descent, and (2) that the descent criterion is computed identically across all five substrate classes in Table 2. The paper’s contribution is to prove the finite components used below, isolate the broader central claims, and state the remaining proof obligations explicitly.

**Formally verified components.** Eight claims in this paper are mechanically verified in Lean 4 (Mathlib v4.30.0-rc2), each axiom-clean (no `sorry`, checked against an explicit axiom allowlist of `Classical.choice`, `Quot.sound`, and `propext`): the finite deterministic factorization biconditional ( $E : H \rightarrow R$ , image-restricted); the *exact finite stochastic factorization* biconditional ( $K : H \rightarrow \text{PMF } R$ , image-restricted, the harder of the two factorization results, closed as separate forward and backward lemmas); the finite kernel-pair non-factorization lemma; the finite-alias Conservative Invariance case; a categorical sheaf-transport statement under suitable equivalence; the Appendix A Case 4  $Z$ -formula evaluations; the Čech-holonomy detection of §5.1 on an arbitrary finite nerve (a closed walk with nonzero holonomy admits no global section, for abelian coefficients, with the cyclic case as a special walk); and the revision-stability of that obstruction under cover refinement (subdividing each edge and redistributing its label leaves the holonomy invariant and nonzero, so no global section exists in any such refinement). Both finite factorization regimes—deterministic and exact-stochastic—are now machine-checked. This last result verifies revision-stability only under refinement of a fixed cover; stability under arbitrary change of measurement language remains argued, not proved. The Lean files are supplied as supplementary material. They are verification receipts for named components, not a proof of the full descent-factorization reduction. That general claim remains argued, not proved, and is the main central reduction still open.

**Principal remaining gap:  $\Omega$  operationalization.** The revision envelope  $\Omega$  must be declared and audited by a claimant before  $Z$  is evaluable. For real physical and computational systems, specifying and publicly justifying  $\Omega$  is open-ended. Practitioners may under-specify, strategically exclude plausible non-conservative revisions, or dispute which expansions count as conservative. This does not affect the theoretical result ( $Z$  is well-defined relative to any declared  $\Omega$ ) but it places the evidential burden on  $\Omega$ ’s completeness and public auditability. The practical next step is therefore an  $\Omega$  audit protocol: a declaration schema, an adversarial challenge process for missing revisions, and a revocation rule when a low-concern verdict rests on an under-specified or strategically narrow envelope. Such a protocol must also refuse a common shortcut: success on seeded omissions qualifies challenger competence but does not bound the probability that adversarial probe classes remain unsurfaced. Residual escape must be bounded under a declared adversary model and public challenge process, or the dossier is non-applicable. The same point applies to independence. Absence of a visible conflict is not a provenance certificate. The dossier must certify the source-class provenance of auditors, red teams, classifiers, panel members, infrastructure providers, and benchmark or data providers. Claimant-originated benefits must be aggregated across proxy and sub-threshold paths. Without that protocol, the framework can state non-applicability cleanly, but a real disclosure dossier remains too easy to game.

**Completed checks.** Internal adversarial review should be treated as a source of candidate objections only. It does not certify the theorem. Evidence for the framework has to come from public proofs, public code, worked systems, or external audit records. Four immediate checks have now been completed: a companion technical report applies the framework to an open-weight transformer (Llama 3.1 8B-Instruct), giving a public-system dossier and a reproducible activation-probe packet for the access/report target; the finite stochastic script checks the exact kernel case; the Boolean recurrent script runs the obstruction calculation on a small transition system; and

the prediction packet states the following falsifiable disclosure-packet prediction: for a fixed text-only, self-hosted, open-weight transformer with no external tools, memory, online learning, or autonomous action loop, access/report use of an input item should be evaluable from public weights, token-level behavior, logits, hidden states, activation probes, causal ablations, prompt interventions, and reproducible inference logs; phenomenal consciousness or moral patienthood should remain non-applicable under that same site unless the claimant supplies a target fiber with decidable equality, a public classifier, a public challenge record, a declared revision envelope, and a provenance certificate for the challenge apparatus. The prediction fails if an auditor supplies those phenomenal-target objects without adding a non-conservative probe, or if the access/report packet cannot produce above-chance behavioral evidence for a system known to use the relevant input.

**Future work.** First, extend the public-system dossier to a deployed system with tool use, memory, and logged deployment state. Second, extend the finite proof from exact kernels to finite-sample statistical decision procedures and finite measurable systems before claiming the fully general version. Third, submit the prediction packet to an external registry or independent auditor before using it as external evidence. Fourth, continue the philosophy-of-mind engagement in §9 by stating exactly which target is at issue: access, report, integration, metacognition, or phenomenal attribution. Fifth, study the reflexive case, in which the observation site is internal to the system  $H$  rather than external to it: a self-monitoring system satisfying P1–P4 cannot in general certify its own global state from its own report, and the kernel-pair obstruction of §4 recurs as a fixed-point constraint on faithful self-representation. We develop this case separately.

## 12 Conclusion

For some hidden-variable systems, consciousness attribution has an identification problem that cannot be removed by additional observations inside the declared site. The case occurs when the target fiber has revision-stable non-effective descent. Section 3 characterizes the systems in this class by four structural properties (P1–P4) that generate necessary non-identification for any substrate that supplies the required site, target fiber, descent semantics, and revision envelope. The revision-stable descent criterion (Theorem 1) states the impossibility under the descent-factorization obligation, and the Proposition (Conservative Invariance) shows that conservative re-presentations of the structural language do not remove the obstruction. Section 5.4 identifies the boundary of the framework: the bridge from P1–P4 to consciousness attribution requires an exogenous commitment or a non-conservative revision of the admissible site. The governance corollary derives from these. When  $Z = 1$ , a low-concern verdict lacks the identification evidence it requires. Any organization issuing such a verdict without first satisfying the six-element disclosure requirement, including a declared revision envelope  $\Omega$ , has not supplied the evidence its verdict needs.

The worked application in §7 is intentionally small. That is the point of it. The framework becomes usable only when the claimant writes down the observation site, target fiber, transition data, and revision envelope for the system at hand. Until those objects are on the page, a low-concern verdict has not been earned; and in a fail-closed audit, the default that follows is restriction, not clearance.

## References

Samson Abramsky and Adam Brandenburger. The sheaf-theoretic structure of non-locality and contextuality. *New Journal of Physics*, 13:113036, 2011. doi: 10.1088/1367-2630/13/11/113036. arXiv:1102.0264.



- Bernard J. Baars. *A Cognitive Theory of Consciousness*. Cambridge University Press, Cambridge, 1988.
- Jonathan Birch. *The Edge of Sentience: Risk and Precaution in Humans, Animals, and AI*. Oxford University Press, Oxford, 2024.
- Ned Block. On a confusion about a function of consciousness. *Behavioral and Brain Sciences*, 18(2):227–247, 1995.
- Patrick Butlin and Theodoros Lappas. Principles for responsible AI consciousness research. *Journal of Artificial Intelligence Research*, 82:1673–1690, 2025. doi: 10.1613/jair.1.17310. arXiv:2501.07290.
- Patrick Butlin, Robert Long, Eric Elmoznino, Yoshua Bengio, Jonathan Birch, Axel Constant, George Deane, Stephen M. Fleming, Chris Frith, Xu Ji, Ryota Kanai, Colin Klein, Grace Lindsay, Matthias Michel, Liad Mudrik, Megan A. K. Peters, Eric Schwitzgebel, Jonathan Simon, and Rufin VanRullen. Consciousness in artificial intelligence: Insights from the science of consciousness. arXiv:2308.08708, 2023.
- Andres Campero, Derek Shiller, Jaan Aru, and Jonathan Simon. Consciousness in artificial intelligence? a framework for classifying objections and constraints. arXiv:2511.16582, 2025.
- Peter Carruthers. *Phenomenal Consciousness: A Naturalistic Theory*. Cambridge University Press, Cambridge, 2000. doi: 10.1017/CBO9780511487491.
- David J. Chalmers. Facing up to the problem of consciousness. *Journal of Consciousness Studies*, 2(3):200–219, 1995.
- Cogitate Consortium, Oscar Ferrante, Urszula Gorska-Klimowska, Simon Henin, Rony Hirschhorn, Aya Khalaf, Alex Lepauvre, et al. Adversarial testing of global neuronal workspace and integrated information theories of consciousness. *Nature*, 642:133–142, 2025. doi: 10.1038/s41586-025-08888-1. Cogitate consortium,  $n = 256$ , multimodal fMRI/MEG/iEEG.
- Stanislas Dehaene and Jean-Pierre Changeux. Experimental and theoretical approaches to conscious processing. *Neuron*, 70(2):200–227, 2011. doi: 10.1016/j.neuron.2011.03.018.
- Daniel C. Dennett. *Consciousness Explained*. Little, Brown and Co., Boston, 1991.
- Keith Frankish. Illusionism as a theory of consciousness. *Journal of Consciousness Studies*, 23(11–12):11–39, 2016.
- Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, et al. The Llama 3 herd of models, 2024.
- Alexander Grothendieck et al. *Revêtements Étales et Groupe Fondamental (SGA 1)*, volume 224 of *Lecture Notes in Mathematics*. Springer, Berlin, 1971. Seminaire de Géométrie Algébrique du Bois-Marie 1960–1961. Descent theory: Exposé VIII–XI.
- Yukio-Pegio Gunji, Yoshihiko Ohzawa, Yuki Tokuyama, Yu Huang, and Kyoko Nakamura. Contextuality as a left adjoint: A categorical generation of orthomodular structure. arXiv:2603.22353, 2026.
- Jake R. Hanson and Sara I. Walker. Integrated information theory and isomorphic feed-forward philosophical zombies. *Entropy*, 21(11):1073, 2019.



- Takao Inoué. On brain as a mathematical manifold: Neural manifolds, sheaf semantics, and leibnizian harmony, 2026.
- Johannes Kleiner and Sean Tull. The mathematical structure of integrated information theory. *Frontiers in Applied Mathematics and Statistics*, 6:602823, 2021. doi: 10.3389/fams.2020.602823. arXiv:2002.07655.
- Joseph Levine. Materialism and qualia: The explanatory gap. *Pacific Philosophical Quarterly*, 64(4):354–361, 1983. doi: 10.1111/j.1468-0114.1983.tb00207.x.
- Robert Long, Jeff Sebo, Patrick Butlin, Kathleen Finlinson, Kyle Fish, Jacqueline Harding, Jacob Pfau, Toni Sims, Jonathan Birch, and David Chalmers. Taking AI welfare seriously. arXiv:2411.00986, 2024.
- Thomas Nagel. What is it like to be a bat? *The Philosophical Review*, 83(4):435–450, 1974. doi: 10.2307/2183914.
- Judea Pearl. *Causality: Models, Reasoning, and Inference*. Cambridge University Press, Cambridge, 2nd edition, 2009.
- Steven Phillips and Naotsugu Tsuchiya. Towards a (meta-)mathematical theory of consciousness: universal (mapping) properties of experience. arXiv:2412.12179, 2024. Revised July 2025.
- Yin Jun Phua. Can we test consciousness theories on AI? ablations, markers, and robustness. arXiv:2512.19155, 2025.
- David M. Rosenthal. *Consciousness and Mind*. Oxford University Press, Oxford, 2005.
- Tony Rost. The sentience readiness index: A preliminary framework for measuring national preparedness for the possibility of artificial sentience. arXiv:2603.01508, 2026.
- Giulio Tononi, Melanie Boly, Marcello Massimini, and Christof Koch. Integrated information theory: From consciousness to its physical substrate. *Nature Reviews Neuroscience*, 17(7):450–461, 2016. doi: 10.1038/nrn.2016.44.
- Naotsugu Tsuchiya, Shigeru Taguchi, and Hayato Saigo. Using category theory to assess the relationship between consciousness and integrated information theory. *Neuroscience Research*, 107:1–7, 2016.
- Ira Wolfson. Informed consent for AI consciousness research: A talmudic framework for graduated protections. *AI and Ethics*, 2026. doi: 10.1007/s43681-025-00852-z. Published online 2025; arXiv:2601.08864.
- Ziheng Zhou, Haiqiang Dai, Bin Ling, Yingnian Wu, and Demetri Terzopoulos. A human-centric framework for debating the ethics of AI consciousness under uncertainty. arXiv:2512.02544, 2025.

## A Falsification Suite

All quantitative predictions are confirmed by the following code, runnable with no dependencies beyond Python’s standard library. The suite tests four claims: (1) the contextuality triangle has  $D = 1$  and  $\check{H}^1 = 1$ ; (2) the kernel-pair case has  $D = 1$  and  $\check{H}^1 = 0$ , demonstrating  $\check{H}^1$ -only recall = 0.5; (3) the identifiable control has  $D = 0$ ; and (4) the revision-stability predictions, namely

that a conservative language expansion leaves  $D$  unchanged and a non-conservative hidden-state-revealing revision removes the obstruction.

```
import itertools

def rank_f2(matrix):
    m = [row[:] for row in matrix]
    if not m:
        return 0
    rows, cols = len(m), len(m[0])
    rank = 0
    for col in range(cols):
        pivot = next(
            (r for r in range(rank, rows) if m[r][col] % 2 == 1), None)
        if pivot is None:
            continue
        m[rank], m[pivot] = m[pivot], m[rank]
        for r in range(rows):
            if r != rank and m[r][col] % 2 == 1:
                m[r] = [(m[r][c] ^ m[rank][c]) for c in range(cols)]
        rank += 1
    return rank

# Case 1: contextuality triangle
CONTEXTS = ["C12", "C23", "C13"]
CONTEXT_STATES = {
    "C12": ("h1", "h2"), "C23": ("h2", "h3"), "C13": ("h1", "h3")}

def overlapping_states(ca, cb):
    return set(CONTEXT_STATES[ca]) & set(CONTEXT_STATES[cb])

def local_value(ctx, state, section):
    return section[CONTEXT_STATES[ctx].index(state)]

def is_globally_consistent(sections):
    for ca, cb in itertools.combinations(CONTEXTS, 2):
        for state in overlapping_states(ca, cb):
            if local_value(ca, state, sections[ca]) != \
               local_value(cb, state, sections[cb]):
                return False
    return True

ODD_PARITY = {
    ctx: [(a,b) for a in (0,1) for b in (0,1) if (a^b)==1]
    for ctx in CONTEXTS}

def global_sections(local_family):
    return [
        {ctx: sec for ctx, sec in zip(CONTEXTS, combo)}
        for combo in itertools.product(
            *[local_family[c] for c in CONTEXTS])
        if is_globally_consistent(
            {ctx: sec for ctx, sec in zip(CONTEXTS, combo)})]

coboundary = [[1,1,0],[0,1,1],[1,0,1]]
triangle_rank = rank_f2(coboundary) # 2
triangle_h1 = 3 - triangle_rank # 1
triangle_gs = len(global_sections(ODD_PARITY)) # 0
triangle_D = 1 if triangle_gs == 0 else 0 # 1
```

```

assert triangle_rank == 2
assert triangle_h1 == 1
assert triangle_gs == 0
assert triangle_D == 1

# Case 2: kernel-pair ( $H^1 = 0$ ,  $D = 1$ )
obs = {"a": "same", "b": "same"}
target = {"a": 0, "b": 1}

def kernel_pair_D(obs, theta):
    for x, y in itertools.combinations(obs, 2):
        if obs[x] == obs[y] and theta[x] != theta[y]:
            return 1
    return 0

kernel_h1 = 0
kernel_D = kernel_pair_D(obs, target) # 1
assert kernel_h1 == 0
assert kernel_D == 1

# Case 3: identifiable control
obs_ctrl = {"a": "obs_a", "b": "obs_b"}
tgt_ctrl = {"a": 0, "b": 1}
control_D = kernel_pair_D(obs_ctrl, tgt_ctrl) # 0
assert control_D == 0

# Detector comparison
total = triangle_D + kernel_D
h1_detected = (1 if triangle_h1 > 0 else 0) + (1 if kernel_h1 > 0 else 0)
h1_recall = h1_detected / total # 0.5
descent_recall = total / total # 1.0

assert h1_recall == 0.5
assert descent_recall == 1.0

```

**Note on Case 4.** This case is a definitional verification, not an independent empirical test. The Conservative Invariance proposition guarantees by category-theoretic argument that adding a Morita-equivalent alias preserves the descent flag. The non-conservative variant adds an observable that distinguishes the two hidden states, which by the definition of the descent obstruction must resolve it. Case 4 is included as a verification benchmark to confirm the formal machinery behaves as the proposition predicts. It does not constitute evidence beyond what the proposition's category-theoretic content already establishes.

```

# Case 4: revision-stability benchmark
# Base: kernel-pair obstruction ( $D_{\text{base}} = 1$ ).
# Conservative revision: add a definitional alias of the existing observable.
# The Proposition (Conservative Invariance) predicts  $D$  unchanged.
# Non-conservative revision: add an observable that distinguishes the hidden
# states.
# This changes  $Y$ ; the obstruction is resolved ( $D = 0$ ).

def compute_D(obs_map, theta_map):
    for x, y in itertools.combinations(obs_map, 2):
        if obs_map[x] == obs_map[y] and theta_map[x] != theta_map[y]:
            return 1
    return 0

```

```

obs_base      = {"a": "obs_X", "b": "obs_X"}
theta_base    = {"a": 0, "b": 1}
D_base        = compute_D(obs_base, theta_base)
assert D_base == 1, "Base kernel-pair must have obstruction"

obs_conservative = {"a": ("obs_X", "obs_X_alias"), "b": ("obs_X", "obs_X_alias")}
theta_conservative = theta_base
D_conservative   = compute_D(obs_conservative, theta_conservative)
assert D_conservative == 1, "Conservative expansion must preserve obstruction"
assert D_conservative - D_base == 0, "Delta D for conservative revision must be 0"

obs_nonconservative = {"a": ("obs_X", "reveals_a"), "b": ("obs_X", "reveals_b")}
theta_nonconservative = theta_base
D_nonconservative    = compute_D(obs_nonconservative, theta_nonconservative)
assert D_nonconservative == 0, "Non-conservative revision must remove obstruction"
assert D_nonconservative - D_base == -1, "Delta D for non-conservative revision
    must be -1"

conservative_overturns      = 0
nonconservative_overturns   = 1
actual_conservative_overturns = 1 - D_conservative
actual_nonconservative_overturns = 1 - D_nonconservative
assert actual_conservative_overturns == conservative_overturns
assert actual_nonconservative_overturns == nonconservative_overturns

n_generators      = 10
m_hidden_states   = 100
revision_subsets  = 2 ** n_generators
hidden_pairs      = m_hidden_states * (m_hidden_states - 1) // 2
total_checks      = revision_subsets * hidden_pairs
assert revision_subsets == 1024
assert hidden_pairs == 4950
assert total_checks == 5_068_800

if __name__ == "__main__":
    print("All assertions passed.")
    print(f"Triangle: rank={triangle_rank}, H1={triangle_h1}, "
          f"global_sections={triangle_gs}, D={triangle_D}")
    print(f"Kernel-pair: H1={kernel_h1}, D={kernel_D}")
    print(f"H1-only recall={h1_recall} | Descent recall={descent_recall}")
    print(f"D_base={D_base}, D_conservative={D_conservative}, "
          f"D_nonconservative={D_nonconservative}")
    print(f"Conservative overturns={actual_conservative_overturns}, "
          f"Non-conservative overturns={actual_nonconservative_overturns}")
    print(f"Revision audit cost (n={n_generators}, m={m_hidden_states}): "
          f"{total_checks:,} pair-checks")

```